

L1b: Time Value of Money, Abstract Assets, and Net Present Value

CHEME 5660 Cornell University

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Objectives for Today

Today we compare **cash flows at different dates**, derive accumulation and discount factors, and use them to value an **abstract asset**.

Today's concepts

- **The time value of money**: cash flows at different dates cannot be combined until they are expressed at the same valuation date.
- **An abstract asset**: a sequence of dated signed cash flows. The same representation covers a bond, a stock, a factory, or a car.
- **Net present value**: discount each signed cash flow to a common valuation date, then add. The sign of the NPV is a measure of the attractiveness of an abstract asset relative to a declared discount rate.

Lectures and Examples

Lecture:

CHEME-5660-L1b-Lecture-TimeValueMoney-Fall-2026.ipynb

Example:

CHEME-5660-L1b-NetPresentValue-Fall-2026-WorkedExample.ipynb

Represent ten years of Tesla Model S ownership as dated cash flows, discount them to time 0, and compute their NPV.

Optional:

CHEME-5660-L1b-Derivation-Discrete-Continuous-Compounding-Fall-2026.ipynb

Measure how far discrete compounding falls short of the fixed-rate continuous limit, and when the estimate of that gap can be trusted.

Link: <https://github.com/varnerlab/CHEME-5660-CourseRepository-Fall-2026.git>

Time Value of Money

A dollar today and a dollar at a future date are **not comparable**. We must express both on a common valuation date using a conversion rule.

Let P be an amount held at time 0, and let y be a quoted **nominal annual yield** used for valuation. With one (discrete) compounding period per year, the amount after one year is given by:

$$F_1 = \underbrace{(1 + y)}_{\mathcal{D}_{1,0}(y;1)} P.$$

For $y > 0$ the factor $\mathcal{D}_{1,0}(y; 1) > 1$ (called the **accumulation factor**) moves time-0 dollars forward in time. It's inverse, $\mathcal{D}_{1,0}^{-1}(y; 1) < 1$ (called the **discount factor**) moves future dollars back in time.

Careful: a later dollar is not *automatically* worth less. That depends on the rate y you declare.

Discrete Compounding

Let $j = 0, 1, 2, \dots$ index the **compounding periods**, let $i_\ell > -1$ be the **effective periodic interest rate** over $\ell \rightarrow \ell + 1$, and collect these as $\mathbf{i} = (i_0, \dots, i_{j-1})$, empty at $j = 0$. An amount P at time 0 becomes F_j at period j :

$$F_j = \underbrace{\left[\prod_{\ell=0}^{j-1} (1 + i_\ell) \right]}_{\mathcal{D}_{j,0}(\mathbf{i})} P, \quad \text{empty product} = 1.$$

For a nominal annual yield y compounded n times per year, every $i_\ell = y/n$:

$$\underbrace{\mathcal{D}_{j,0}(y; n) = \left(1 + \frac{y}{n}\right)^j}_{\text{accumulation}}, \quad \underbrace{\mathcal{D}_{j,0}^{-1}(y; n) = \left(1 + \frac{y}{n}\right)^{-j}}_{\text{discount}}.$$

Assumes: constant y/n , end-of-period compounding, reinvested interest, $1 + y/n > 0$.

Careful: a flat y is an *assumption*. In L2b, one security's yield gives way to a curve of spot rates.

Continuous Compounding

Let $g(t)$ be a **continuously compounded growth rate**, carrying units of inverse time, and let T be the horizon in years. Over $0 \rightarrow T$:

$$\mathcal{D}_{T,0}(g) = \exp \left(\int_0^T g(u) du \right), \quad \mathcal{D}_{T,0}^{-1}(g) = \exp \left(- \int_0^T g(u) du \right),$$

which for **constant** g collapses to:

$$\underbrace{\mathcal{D}_{T,0}(g) = e^{gT}}_{\text{accumulation}}, \quad \underbrace{\mathcal{D}_{T,0}^{-1}(g) = e^{-gT}}_{\text{discount}}.$$

A nominal annual yield y compounded n times per year fits this form. Setting $g = g_y$ with $g_y := n \log(1 + y/n)$ makes $(1 + y/n)^{nT} = e^{g_y T}$ hold exactly at every finite n : y and its continuous equivalent g_y are **two representations of one accumulation rule**.

Careful: at finite n , $y \neq g_y$, so y does not belong in e^{yT} . Only as $n \rightarrow \infty$ does $g_y \rightarrow y$.

Who Sets the Valuation Rate?

The valuation rate is **not universal**. It depends on the cash flow, market convention, horizon, and the risks being modeled: no single policy rate prices every asset.

THE FEDERAL RESERVE

- The Fed conducts **monetary policy** for maximum employment, stable prices, and moderate long-term rates, **supervises banks**, and is the **lender of last resort**.
- The Federal Open Market Committee (**FOMC**) sets a target range for the federal funds rate, the overnight rate on unsecured reserve balances traded between eligible institutions.
- Other market rates, e.g., Treasury yields and mortgage rates, respond to policy and market conditions, but are **not mechanically pegged** to the federal funds rate.

Link: [FOMC calendar and meeting materials](#)

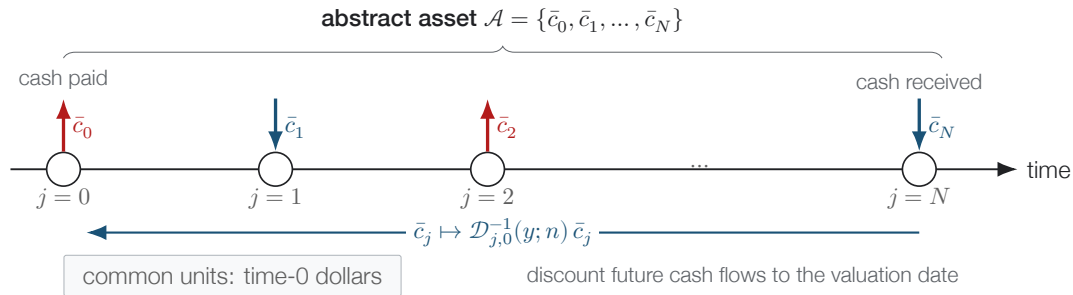
What Valuation Rate Should I Use?

There is no universal answer: y is a **declared model input**, matched to the cash flow by currency, horizon, and any stated adjustment for credit, liquidity, or risk.

Term	Symbol	Meaning here
Interest rate	i_ℓ	A rate paid or received for borrowing or lending, such as the federal funds rate or a mortgage rate. Those are annual quotes; the model uses $i_\ell = y/n$ per period.
Yield	y	A rate tied to the price and cash flows of an investment, such as a bond yield.
Discount rate or valuation rate	y	The rate a model uses to convert future cash flows to present value.
Continuous equivalent	g_y	The same rate written in continuous form, $g_y = n \log(1 + y/n)$.

Careful: the same y fills rows 2 and 3: one number, two roles. From L2a, g is an asset's own measured growth rate.

An Abstract Asset



Net Cash Flows

An abstract asset is a **sequence of dated cash flows** in one currency, one at each compounding date $j = 0, 1, \dots, N$, so N is the final index and there are $N + 1$ dates. At date j , let $\mathbf{c}_j \in \mathbb{R}_{\geq 0}^{m_j}$ hold the m_j nonnegative cash flow magnitudes and $\boldsymbol{\nu}_j \in \{-1, +1\}^{m_j}$ their directions, $+1$ an inflow and -1 an outflow. The **signed net cash flow** is their inner product:

$$\bar{c}_j = \langle \mathbf{c}_j, \boldsymbol{\nu}_j \rangle = \sum_{k=1}^{m_j} c_{j,k} \nu_{j,k}.$$

The undiscounted total $\sum_{j=0}^N \bar{c}_j$ is **not** a time-0 value: its terms are dollars from different dates.

Next: discount every term to a common date, and the result is net present value.

Net Present Value

Take a nominal annual yield y compounded n times per year as the declared valuation rate. Discount every signed net cash flow $\bar{c}_j$ to the valuation date with $\mathcal{D}_{j,0}^{-1}(y; n)$, then add. The net present value at time 0 is given by:

$$\text{NPV}_0(y) = \sum_{j=0}^N \mathcal{D}_{j,0}^{-1}(y; n) \bar{c}_j = \underbrace{\langle \mathcal{D}_{\star,0}^{-1}(y; n), \bar{\mathbf{c}} \rangle}_{\text{time-0 dollars}}.$$

The $\star$ runs over every cash-flow index, so $\mathcal{D}_{\star,0}^{-1}(y; n)$ and $\bar{\mathbf{c}}$ are ordered $(N+1)$ -vectors and NPV is a single scalar product, which is exactly how we compute it in code.

Worked example: ten years of Model S ownership, $\text{NPV}_0 \approx -\$121,484$

Reading the Sign of NPV

The sign of $NPV_0(y)$ compares the modeled cash flows against the **rate you declared**. It is a valuation statement, not a promise of accounting profit or realized return.

- **Positive:** in time-0 dollars, receipts exceed payments, so at the declared y the asset is worth more than its price.
- **Negative:** receipts fall short of payments, so at the declared y the asset is worth less than its price.
- **Zero:** receipts and payments agree, so the asset is worth exactly its price. This is how L2a prices Treasury bills, notes, and bonds.

Change y and the sign can change: it describes this cash-flow model at this rate, not the asset in general.

Careful: a positive NPV is not a claim that an asset is risk-free or universally attractive.

Summary

This lecture introduced time-value-of-money calculations and the representation of investments as dated cash flows.

Key takeaways

- **Dated dollars must be compared at a common valuation date.** A forward accumulation factor carries value forward and its inverse discounts it back, so cash flows can be added only once they share a date.
- **Rate conventions must be stated.** A nominal yield quoted with a compounding frequency and its continuously compounded equivalent describe one accumulation rule, so the convention travels with the number.
- **NPV values an asset relative to a declared rate.** The sign of the result is a statement about the cash-flow model and the discount rate you declared.

These ideas provide the valuation framework used throughout the course.